Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/05/2008
PI Name:	Jin, J.-P. M.D., Ph.D.		
Project Title:	Adaptation Of Myocardial Contractile Proteins In Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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City:	Evanston	**State:**	IL
Zip Code:	60201-1718	**Congressional District:**	9
Comments:	Former affiliation: Case Western Reserve University; moved in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	07/01/2004	**End Date:**	01/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received extension to 1/31/2009 per A. Chu/ARC (6/2008) NOTE: NCE to 6/30/2008 per A. Chu (4/07) NOTE: NCE to 6/30/07 per A. Chu (7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK62G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), an essential protein in the Ca2+-regulatory system of cardiac muscle. This novel structural regulation of cTnI removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA). Beta-adrenergic-regulated PKA phosphorylation of cTnI reduces the sensitivity of cardiac muscle to Ca2+. Therefore, the up-regulation of N-terminal truncated cTnI may provide a compensation for cardiac function against the microgravity-induced reduction of contractility. This research proposal describes ground based studies to investigate the role of the proteolytic regulation of cTnI in adaptation to microgravity for potential use in the prevention and treatment of cardiovascular deconditioning during and after space flight. We plan to study the functional changes in cTnI caused by the N-terminal truncation and the effect on cardiac muscle contraction in simulated microgravity. Conformation and function of the N-terminal truncated cTnI will be investigated in vitro and in the heart of transgenic mice. We shall examine the effect of the removal of PKA phosphorylation sites from cTnI on beta-adrenergic regulation during cardiac adaptation in simulated microgravity. We shall also investigate the regulatory mechanisms for this proteolytic modification of cTnI during adaptation to microgravity. We shall focus on the role of calpain that is a family of regulatory proteases and known to function in the degradation of troponin subunits. The role of PKA and PKC phosphorylation of cTnI in the regulation of sensitivity to calpain cleavage will be characterized. This research will provide new information for understanding cardiac adaptation in microgravity and protecting cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our study will contribute to both space life science and human life quality on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our discovery that the N-terminal modification of cardiac troponin I enhances cardiac function has lead to an approach to the development of new drugs for the treatment of heart failure that is the most important death cause for cardiovascular diseases.
Task Progress:	Period covered by the report 06/30/07-05/31/08 Project Summary: In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), an essential protein in the Ca(2+)-regulatory system of cardiac muscle. This novel structural regulation of cTnI removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA). Beta-adrenergic-regulated PKA phosphorylation of cTnI reduces the sensitivity of cardiac muscle to Ca(2+). Therefore, the up-regulation of N-terminal truncated cTnI may provide a compensation for cardiac function against the microgravity-induced reduction of contractility. This research project is a ground-based study to investigate the role of the proteolytic regulation of cTnI in adaptation to microgravity for potential use in the prevention and treatment of cardiovascular deconditioning during and after space flight. We are studying the functional changes in cTnI caused by the N-terminal truncation and the effect on cardiac muscle contraction in simulated microgravity. Conformation and function of the N-terminal truncated cTnI are investigated in vitro and in the heart of transgenic mice. We are examining the effect of the removal of PKA phosphorylation sites from cTnI on beta-adrenergic regulation during cardiac adaptation in simulated microgravity. We are also investigating the regulatory mechanisms for this proteolytic modification of cTnI during adaptation to microgravity. We focus on the role of calpain that is a family of regulatory proteases and known to function in the degradation of troponin subunits. The role of PKA and PKC phosphorylation of cTnI in the regulation of sensitivity to calpain cleavage will be characterized. This research will provide new information for understanding cardiac adaptation in microgravity and protecting cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our study will contribute to both space life science and human life quality on Earth. Progress in the reporting period 1) A review article including work from this study has been published: Jin, J.-P., Zhang, Z., and Bautista, J.A. (2008) Isoform diversity, regulation and functional adaptations of troponin and calponin. Crit. Rev. in Eukar Gene Expr. 93-124. 2) Three research papers from work done in this study have been submitted for publication: Feng, H.-Z., Chen, M., Weinstein, L.S. and Jin, J.-P. Removal of the N-terminal extension of cardiac troponin I as a functional compensation for myocardial ß-adrenergic deficiency. Submitted. You, B., Yan, G., Zhang, Z., Yan, L., Chen, W., Jin, J.-P., and Sun, J. Phosphorylation of Cardiac Troponin I by Mammalian Sterile20-Like Kinase 1. Chong, S.M., and Jin, J.-P. To investigate protein evolution by analyzing default structural states. Submitted. 3) A research paper from work in this study is currently under revision for publication: Bradley K. McConnell, Zoran Popovic, Niladri Mal, James Bautista; Farhad Forudi, Raul Schwartzman, J.-P. Jin; Marc Penn; Meredith Bond; Targeted Disruption of PKA / AKAP Interaction: Effects on Cardiac Contractility, PKA Phosphorylation, and TnI Proteolysis in rat Hearts in vivo. 4) Lecture Presentations: 08/2007 Washington State University. Title: Regulation of troponin function by restricted proteolysis. 09/2007 The 36th European Muscle Conference, Stockholm, Sweden. Title: Proteolytic modification of cardiac troponin T and troponin I in functional adaptation. 10/2007 University of North Texas. Title: Tension regulation and function of h2-calponin. 11/2007 Institute of Biophysics, Chinese Academy of Sciences. Title: Troponin function and myocardial Contractility. 04/2008 Symposium of NPU Special Environmental Biophysical Institute, Xi'an, China. Title: Physiological adaptation in health and diseases.

	04/2008 University of Maryland School of Medicine. Title: Troponin in cardiac function. 05/2008 Conference on Thick and Thin Filament Regulation in Striated Muscle, Madison, WI. Title: Regulation of Troponin T Function by Restricted Proteolysis. Future research plan We will continue a) searching for chemical compounds that can mimic the positive effect of cardiac troponin I N-terminal truncation on the heart for use as a countermeasure to the decreased cardiac function in astronauts during long space flight; b) testing the effects of the N-terminal truncated cardiac troponin I on compensating for the lost beta-adrenergic potential in transgenic mouse hearts; and c) investigating the mechanism that regulates the N-terminal truncation of cardiac troponin I for a potential target in the treatment of cardiac inefficiency of astronauts and heart failure patients on Earth. There has been no significant change in the research plan.
Bibliography Type:	Description: (Last Updated: 04/20/2009)
Articles in Peer-reviewed Journals	Jin JP, Zhang Z, Bautista JA. "Isoform diversity, regulation, and functional adaptation of troponin and calponin." Crit Rev Eukaryot Gene Expr. 2008;18(2):93-124. Review. PMID: 18304026 , Mar-2008